Evan Pete Walsh

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

(1) Define the following: (a) Semi-algebra, (b) algebra, (c) σ-algebra, (d) measurable space, (e) measure, (f) finite measure and probability, (g) σ-finite measure, (h) measure space and probability space, (i) measurable transformation, (j) random variable.

($4\times 10 = 40$ points)

(2) Let $\mathcal{F}$ be a σ-algebra of subsets of $\Omega \neq \emptyset$. Take $\Omega_0 \subset \Omega$ such that $\boxed{\Omega_0 \notin \mathcal{F}.}$ Define $\mathcal{F}_0 = \{F \cap \Omega_0 : F \in \mathcal{F}\}$. Show that $\mathcal{F}_0$ is a σ-algebra of subsets of Ω_0. $\Omega_0 \cap A^c$

(15 points)

$\Omega \cup A$

(3) (a) Give an example of a functions f that is **not** $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R})\rangle$-measurable.
(b) Give examples of two (distinct) functions f and g that are **not** $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R})\rangle$-measurable, but $f+g$ is $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R})\rangle$-measurable.
(c) Let $\Omega_1 = \mathbb{R}$ and $\mathcal{F}_1 = \{\emptyset, (-\infty, 5], (5, \infty), \mathbb{R}\}$ and let $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Let $T : \Omega_1 \to \Omega_2$ be defined as

$$T(x) = \begin{cases} 5, & \text{if } x \leq 0 \\ 0, & \text{if } x > 0 \end{cases}$$

Is T $\langle \mathcal{F}_1, \mathcal{F}_2\rangle$-measurable? Justify your answer.

($2 \times 10 = 20$ points)

$\Omega \setminus A$

(4) Let $\Omega \neq \emptyset$ and $\mu^* : \mathcal{P}(\Omega) \to [0, \infty]$ be a function ($\mathcal{P}(\Omega)$ is the power set of Ω).
(a) Define what properties μ^* need to satisfy to be an *outer-measure*.
(b) If μ^* is an *outer-measure*, define what it means for a set $A \subset \Omega$ to be μ^*-measurable.
(c) Show that if an outer measure μ^* is also *finitely-additive*, then it is actually a measure on all subsets of Ω (i.e. on $(\Omega, \mathcal{P}(\Omega))$). State any results used in your argument. *[Hint: start by finding all μ^*-measurable sets]*

($3 \times 10 = 30$ points)

$0 \quad \in A$
$\in A^c$

$1 \in A$ $\quad 1$
$0 \in A^c$

$M_{\mu^*} = \{A \subseteq \Omega : \mu^*(x) = \mu^*($

$\Omega_0 \setminus A = (\Omega \setminus F) \cap \Omega$

Let ~~[illegible]~~ $A \subseteq \Omega$, $X \subseteq \Omega$

$f+g = 1$

$\bigcup F_n$